MIL-P-85511A(AS)
5 JUNE 1987
SUPERSEDING
MIL-P-85511
5 March 1982

MILITARY SPECIFICATION

PANEL ASSEMBLY, TARGET, BANNER, AERIAL TDU-32 SERIES

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the TDU-32 series aerial banner target panel assembly, referred to herein as the panel assembly.

1.2 Classification. The panel assembly covered by this specification consists of the following types:

Type I. Radar non-reflective panel assembly, used on the TDU-32/B Aerial Banner Target.

Type II. Radar reflective panel assembly, used on the TDU-32A/B Aerial Banner Target.

Type III. Laser retroreflective panel assembly, used on the TDU-32 B/B Aerial Banner Target.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, New Jersey 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1550

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:59Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issues of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods
FED-STD-595	Colors

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
DOD-STD-2101	Classification of Characteristics

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

DL 966AS8000 Rev K	Target, Banner, Aerial, TDU-32 Series

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:59Z
Check the source to verify that this is the current version before use.

NAVAIR 28-10A-501 Aerial Targets and Associated Equipment and Service Instruction

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.1.1 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), the first article sample shall consist of three of each type of panel assemblies. The first article sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production.

3.2. Design and construction. The panel assembly shall be manufactured and assembled in accordance with the drawings and other documents listed on DL 966AS8000, and as specified herein.

3.2.1 Material characteristics and panel assembly. The material used in the fabrication of the panel assembly shall be selected by the manufacturer, and shall meet the requirements of 3.3 and 3.3.1. The finished panel assembly shall be continuous with no vertical splice permitted across the width of the panel.

3.2.1.1 Type I Panel Assembly. The material used in the fabrication of the Type I panel assembly shall be transparent to radio frequency (RF) signals.

3.2.1.2 Type II Panel Assembly. The material used in the fabrication of the Type II panel assembly shall reflect RF signals.

3.2.1.2.1 Radar reflectivity. When illuminated with horizontal or vertical polarized radar energy of 10.0 gigahertz (GHz) frequency, a one square foot sample of material from the Type II panel assembly which has withstood three missions specified in 3.3, shall reflect no less than 90 percent of the radiated energy when radiated normal to the plane of the sample. The above sample shall be representative of the average condition of the panel assembly.

3.2.1.3 Type III Panel Assembly. The material used in the fabrication of the Type III panel assembly shall reflect RF signals and retroreflect laser radiation.

Source: https://assist.dla.mil -- Downloaded: 20261O-23T10:59Z
Check the source to verify that this is the current version before use.

3.2.1.3.1 Laser Retroflectivity. When illuminated with infrared radiation at 904 nanometers from a gallium arsenide laser at normal incidence, a sample of the coated Type III panel assembly which has withstood three missions specified in 3.3, shall exhibit a retroreflectivity of not less than 1.9 (watt/steradian)/watt. The above sample shall be representative of the average condition of the tow banner. The radar reflectivity requirement of 3.2.1.2.1 shall also apply to the Type III panel assembly.

3.2.1.4 Abrasion. The panel assembly, when subjected to abrasion or burning during ground-drag launched take off, shall meet the performance requirement specified herein.

3.2.1.5 Fraying and ripping. Fraying or ripping of the trailing edge of the panel assembly shall be not greater than two feet for each hour of flight.

3.2.1.6 Bullet marking. The material used in the construction of the panel assembly, when perforated by ammunition which has been tipped with colored coating compounds, shall retain visible traces of the compound used to provide a means of scoring projectile hits. The color coating compounds in Table I shall be used.

TABLE I. Colors of coating compounds.

COLOR	NATIONAL STOCK NUMBER (NSN)
Blue	8010-00-899-3173
Black	8010-00-899-3174
Green	8010-00-899-3175
Yellow	8010-00-899-3176
Purple	8010-00-899-3177
Red	8010-00-899-3178
Brown	8010-00-899-3179

3.2.1.6.1 Bullet perforation runs. The material used in the construction of the panel assembly shall not exhibit runs as a result of bullet perforation.

3.2.1.7 Visibility. In order to provide the maximum in-flight visibility, the finished panel material shall be painted on both sides with white vinyl conforming to FED-STD-595 Color number 27886 insuring 100 percent coverage of all exposed yarn surfaces. Painting shall be completed by the commercially available dipping process and shall exhibit adequate adhesion for the intended use.

3.3 Performance. The panel assembly shall withstand three missions. Each mission shall consist of a ground-drag launch, a continuous flight of not less than 1.5 hours from sea level to 30,000 feet altitude at a speed of 250 knots indicated air speed (KIAS), and release of the target from the aircraft at an altitude not greater than 500 feet and a velocity not greater than 175 KIAS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:59Z
Check the source to verify that this is the current version before use.

3.3.1 Flight characteristics. The panel assembly, when attached to the tow bar and bridle assembly in accordance with DL 966AS8000 and flown in accordance with 3.3, shall not cause deformations of the tow bar or exhibit panel distortion so as to deviate from planar upright flight.

3.4 Environmental.

3.4.1 Temperature. The panel assembly shall perform as specified herein from -48°F to +140°F (-44°C to +60°C).

3.5 Panel assembly marking. Each panel assembly shall be marked in accordance with DL 966AS8000.

3.5.1 Marking legibility. The marking of each panel assembly shall remain legible after the three missions of 3.3.

3.6 Workmanship. There shall be no imperfections in the panel assembly which would prevent the requirements specified herein from being met.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2. Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:59Z
Check the source to verify that this is the current version before use.

4.3 Inspection conditions. Prior to testing, all samples shall be held at standard atmospheric conditions as described in FED-STD-191, Section 4, "Atmospheric Conditions for Testing", until equilibrium has been attained and shall be tested at standard atmospheric conditions.

4.4 First article inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the first article inspection shall consist of all inspections in Table II during contractor's inspection, and Table III during Government flight evaluation. The contractor shall perform all the inspections of Table II prior to forwarding the first article, at his own expense, to the testing activity designated in the contract or purchase order (see 6.2.1), for flight test evaluation. Acceptance of the first article shall be based upon successfully passing the inspection of Table II and Table III.

TABLE II. Contractor's first article inspection.

Inspection	Requirement paragraph	Test method paragraph
Design and construction	3.2	4.5, 4.5.1, 4.5.1.1, 4.5.1.2.1
Material	3.2.1	4.8
Product marking	3.5	4.5.1.1

TABLE III. Flight evaluation first article inspection.

Inspection	Requirement paragraph	Test method paragraph
Radar reflectivity	3.2.1.2.1	4.8.1
Laser Retroreflectivity	3.2.1.3.1	4.8.2
Abrasion	3.2.1.4	4.8.3
Fraying or ripping	3.2.1.5	4.8.4
Bullet marking	3.2.1.6	4.8.5, 4.8.6
Visibility	3.2.1.7	4.8.7
Performance	3.3	4.7
Flight characteristics	3.3.1	4.7

4.5 Quality conformance inspection. Unless otherwise specified herein, the contractor shall perform all quality conformance inspections. The contractor shall conduct and show evidence of a detailed visual examination to demonstrate that the panel assembly has been manufactured and assembled in accordance with DL 966AS8000 and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:59Z
Check the source to verify that this is the current version before use.

4.5.1 Examination of the panel assembly.

4.5.1.1 Complete assembly. Each panel assembly shall be examined to determine that workmanship requirements have been met and that the complete assembly conforms to DL 966AS8000 and as specified herein with respect to the following:

a. Completeness of detail.

b. Identification properly positioned and marked.

c. Material characteristics.

4.5.1.2 Classification of characteristics. The panel assembly characteristics to be verified by the inspections specified herein are classified critical, major, or minor in accordance with DOD-STD-2101. This classification is accomplished by insertion of a classification of characteristics symbol and serial number to the left of the number of the paragraph to which it applies. Critical characteristics are identified by an upper case letter "C" followed by an arabic numeral 1 through 99. Major characteristics are identified by an upper case letter "M" followed by an arabic numeral 101 and up. Minor characteristics are those characteristics which are not identified by any classification of characteristics symbol. Defective critical characteristics are critical defects. Defective major characteristics are major defects. Defective minor characteristics are minor defects.

4.5.1.2.1 Classification of drawing requirements. The drawing requirements for DL 966AS8000 shall be inspected on the basis of inspections in accordance with 4.5.1.2 and Table IV.

4.5.1.3 Lot sampling. Lot sampling for quality conformance inspection shall be in accordance with MIL-STD-105, Inspection Level II Table I. Single sampling plans for normal inspection (master table) Table IIA. The acceptance quality level (AQL) is 4.0 for major characteristics and 25.0 for minor characteristics for the lot size specified in the contract or purchase order (see 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:59Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of characteristics.

Classification	Defect
Critical	None
Major	
(M101)	Verify that the shaded areas specified on the drawing are within the dimensional tolerances specified and colored florescent orange conforming to FED-STD-595, Color no. 38903.
(M102)	Verify that the striped areas specified on the drawing are within the dimensional tolerances specified and colored silver-gray conforming to 3M brand paint code 7210 or close equivalent thereof.
(M103)	Verify that the dimensions of the loop are 5.0 inches.
(M104)	Verify that the diameter of the bull's-eye is 48.0 ± 1.0 inches and is located within the dimensional tolerances specified.
(M105)	Verify that the width of the panel is 90.0 ± 2.0 inches.

4.6 Inspection and test procedures. The contractor and the testing activity performing the flight evaluation testing shall prepare test procedures in accordance with MIL-STD-831. The right is reserved by the procuring activity or Government representative to modify or require additional inspection or tests necessary to determine compliance with the requirements of this specification.

4.7 Flight inspections. The panel assembly shall be assembled to the tow bar and bridle assembly in accordance with Drawing 966AS8000 and ground launched in accordance with NAVAIR 28-10A-501. After flight and ground recovery, each panel assembly shall be subjected to the inspections of 4.8 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:59Z
Check the source to verify that this is the current version before use.

4.8 Material inspection.

4.8.1 Radar reflectivity. Radar reflectivity of the panel assembly shall be measured after completion of three missions specified in 3.3 to determine conformance to 3.2.1.2.1 for Type II and Type III panels.

4.8.2 Laser Retroreflectivity. Laser retroreflectivity of the panel assembly shall be measured after completion of three missions specified in 3.3 to determine conformance to 3.2.1.3.1 for Type III panels.

4.8.3 Abrasion. After one mission specified in 3.3, the panel assembly shall be examined to determine conformance to 3.2.1.4.

4.8.4 Fraying or ripping. After one mission specified in 3.1, the panel assembly shall be examined to determine conformance to 3.2.1.5.

4.8.5 Bullet marking. The panel assembly shall be perforated by 18 rounds of either .50 caliber or 20 millimeter projectiles tipped with coating compounds specified in Table I. The panel assembly material adjacent to the projectile holes shall bear evidence of coating compound and each color shall be distinguishable from any other color.

4.8.6 Bullet perforation runs. The panel assembly material shall be examined to determine conformance to 3.2.1.6.1.

4.8.7 Visibility. The panel assembly material shall be examined to determine conformance to 3.2.1.7.

4.9 Environmental.

4.9.1 Temperature. The temperatures encountered during the flight inspections specified in 4.7 shall be the only test required to determine conformance to 3.4.1.

5. PACKAGING.

5.1 Packaging and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), the panel assembly shall be packaged and packed in accordance with MIL-STD-794, Level A.

5.2 Marking for shipment and storage. Marking for shipment and storage shall be in accordance with MIL-STD-129.

5.3 Special markings. The following information shall be placed on each package, with one inch high red letters, so as to be conspicuous from all sides and not interfere with any other markings:

"DO NOT OPEN WITH SHARP INSTRUMENT"

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:59Z
Check the source to verify that this is the current version before use.

6. NOTES.

6.1 Intended use. The panel assembly is intended for use as part of a highly visible, aircraft towed target for gunnery training exercises, used on the TDU-32 series Aerial Banner Target.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity and type for first article sample (see 3.1.1).

c. Responsibility for inspection (see 4.1).

d. First article inspection, if different than specified (see 4.4).

e. Designated testing activity for flight evaluation (see 4.4).

f. Lot size (see 4.5.1.3).

g. Level of packaging and packing, if different than specified (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID no.	Option
4.4	Report, First Article Test	UDI-T-23790	----
4.6	First Article Inspection Procedures	DI-T-4901	----

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5010.12L AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:59Z
Check the source to verify that this is the current version before use.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of three of each type of panel assemblies. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Subject term (key word) listing. Subject term key words for identification during retrieval searches are as follows:

Aerial Gunnery Banner
Aerial Tow Target, Banner, Panel Assembly
Gunnery Banner
Target, Banner, Panel Assembly
Tow Target, Banner, Panel Assembly

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - AS

(Project 1550-N005)

U.S. GOVERNMENT PRINTING OFFICE: 1987 - 704-036/73566

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:59Z
Check the source to verify that this is the current version before use.